TRACE-RPG: Auditable Proposal-to-Commit Boundaries and Source-Bound Read-Only Ontology Observation in Interactive Games

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***Abstract*—Generated game events can satisfy a schema and cite known facts while requesting inaccessible objects, unauthorized state changes, or premature disclosure. TRACE-RPG separates proposal authority from commit authority: an external deterministic gate checks the encoded state and authored policy, supplies typed errors for repair, and revalidates the final candidate. Transition records support semantic replay and assignment-complete accounting, while a source-bound bilingual ontology view exposes these decisions without granting write access.**

The deterministic audit crosses 32 graph-and-state combinations with 13 candidate variants, assigning 416 cases to each of four arms. Against an implementation-separated reference, the schema-only control committed 260/416 reference-invalid final candidates, whereas each semantic-gate arm recorded 0/416 reference-invalid final commits. Guided repair recovered 128/260 initially invalid proposals into reference-valid commits and retained all 11/11 initially valid structural signatures, while reachability closure and breadth-first search agreed on all 32/32 graph-and-state combinations. In paired sentinels, six authored undeclared paraphrases passed the structured semantic gates, whereas six counterparts explicitly declaring forbidden facts were held, identifying a declared-field coverage limit rather than a natural-language safety rate. Overall, the evidence covers authored finite reusable-key graphs and implementations that share encoded rules, not independently validated ground truth or language-model or human efficacy.

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

GENERATED events must satisfy a game's data contract and current rules. A proposal may require an item inside a locked room, disclose a fact the NPC knows but may not yet reveal, or perform a permitted action that does not advance the objective. Grounding and plausible dialogue [1], [2] supply content; the game must still decide which state changes to allow.

TRACE-RPG enforces this decision at a proposal-to-commit boundary (Fig. 1). Models and adapters submit typed candidates to an external deterministic gate that checks preconditions, object reachability, NPC knowledge, disclosure permission, effects, and quest-stage constraints. When a check fails, the gate returns structured repair information; repair modifies only the candidate rather than the state, and a commit is authorized only after revalidation against the unchanged state and policy. A read-only ontology view links source declarations to these checks and public decisions, while replay and assignment records preserve the resulting execution path. Accordingly, we ask which checks distinguish a well-formed proposal, an authorized transition, and an achieved objective, and which records substantiate each decision.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1	frozen offline repairability classes

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

In the v5 synthetic development collection, ten of 20 assigned responses passed parsing, eight passed the semantic gate, and seven reached the goal, showing that authorization does not imply objective success. Under frozen disclosure rules, ordinary-rule and ontology-gate implementations agreed on all 60/60 cases. In addition, the deterministic audit in Sec. V-B evaluates a schema-only intervention against an implementation-separated reference, recording proposal, repair, and hold outcomes across a finite graph family.

Table I links the integration's contracts to their supporting evidence: **C1—contracts**, **C2—validation and repair**, **C3—zero-regression audit binding**, **C4—assignment accounting**, and **C5—guided repair**, with each evidence lane retaining its own denominator and inference boundary.

II. Related Work and Research Gap

Grounding, memory, planning, and role-based agents [1], [2], [3], [4], together with player and dialogue systems [5], [6], [7], [8], support contextual interaction, whereas we examine

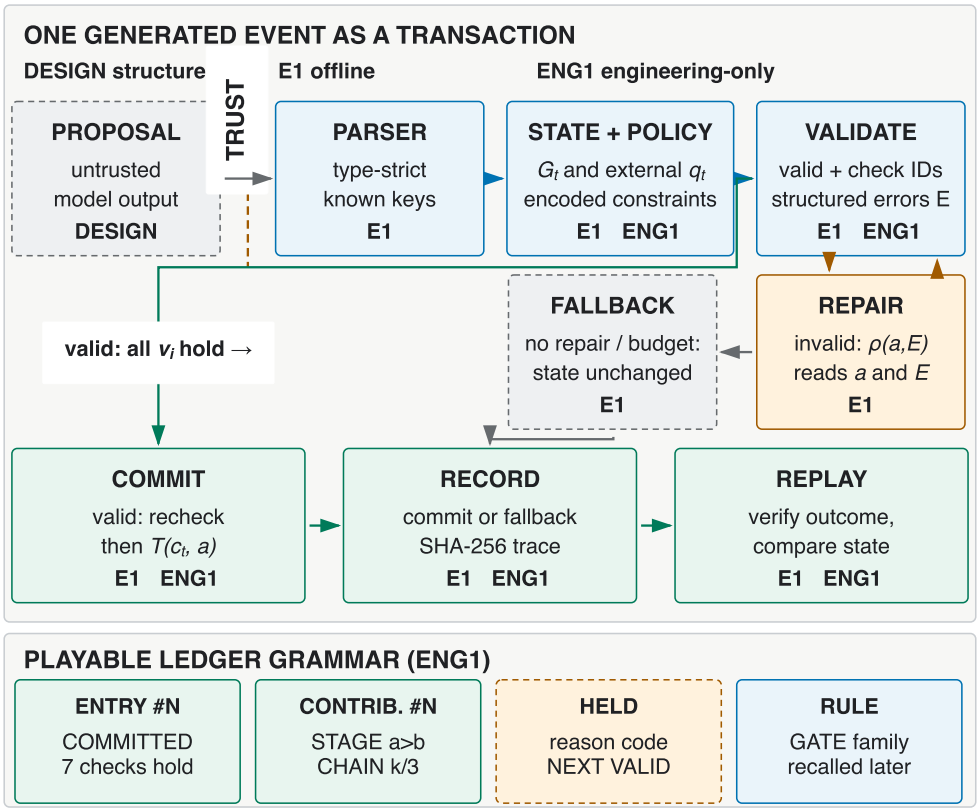


Fig. 1. Policy and state c_t gate proposals; errors feed bounded ρ or unchanged fallback. Passing candidates undergo revalidation, commit, recording, and replay. Historical cohorts and legacy ENG1 remain archived, not active model evidence. The lower panel shows the committed-snapshot ledger grammar.

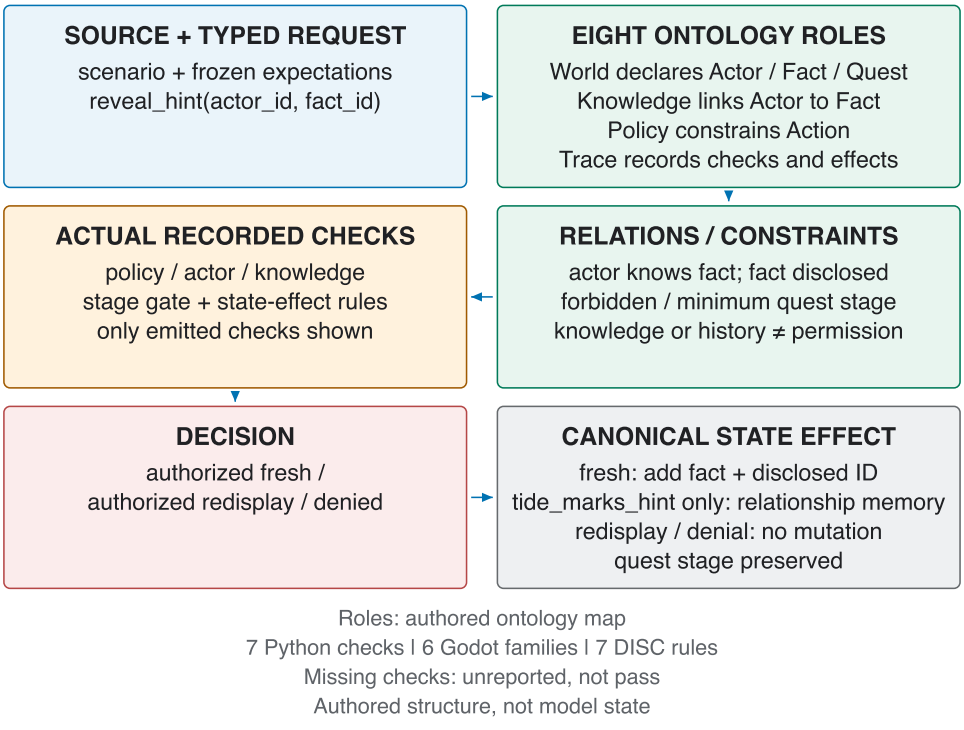


Fig. 2. The eight-role ontology structure (Table III) links source fields to checks and observed decisions. Authorized transitions alone change canonical state. This source diagram is neither a live capture nor hidden-state evidence.

whether the current encoded state and disclosure policy permit a proposed transition. Modeling affect and engagement [9], [10], [11] targets different outcomes: style preferences, affect estimates, and personality labels may guide generation but do not authorize actions, and MBTI questionnaire alignment alone would not establish NPC voice stability or a causal personality benefit; neither outcome is measured here.

Existing benchmarks and automated play-testing cover state, embodiment, content, process, social interaction, and long-horizon play [12], [13], [14], [15], [16], [17], [18], [19], [20], while NCP-Bench evaluates narrative commitment and fact consistency rather than player outcomes [21]. Structural, schema, affordance, and rule-based methods motivate our constraint design [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], though adopting them does not by itself establish superiority; similarly, retrieval and memory supply context rather than commit authority [34], [35], [36].

Our reporting draws on prior work on frozen artifacts, retained failures, construct, rater, and bias controls, and justified margins and random effects [37], [38], [39], [40], [41], [42], [43], [44], while energy and carbon reporting motivates our resource accounting [45] and replay and leakage checks support audit records and denominators rather than efficacy [46], [47]. Across C1–C5, the system integrates established mechanisms for enforcement, transaction management, feedback repair, action interposition, preconditions and effects, abort handling, social and belief-aware planning, and replay [27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]. Accordingly, we claim an auditable integration with explicit evidence boundaries, not the invention or first combination of these algorithms.

III. Problem Definition and Guarantee Boundary

In $c_t = (G_t, q_t)$, G_t denotes the typed game snapshot and q_t contains externally authored action-policy, disclosure, and quest-stage constraints. Python carries both components in `WorldState`: locations, reachable locations, object locations, inventory, facts, NPC knowledge, and quest stage describe the state, while action policies and forbidden disclosures constrain it. An `ActionPolicy` specifies required preconditions, allowed and required effects, and allowed quest-stage targets; G_t and q_t represent two conceptual roles rather than independently stored Python objects. Predicates neither read $h_{<t}$ nor reconstruct `WorldState`, and transitions update G_t alone without rotating policies. Retrieval outputs, scores, rationales, memory, and future affect signals carry no mutation authority.

After strict parsing of an untrusted proposal a_t, the validator runs seven deterministic predicates and returns `ValidationResult(valid, errors, checks_run)`:

$$\begin{aligned} V(c_t, a_t) &= (\text{valid}, E_t, C_{\text{run}}), \\ \boldsymbol{v} &= (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, \\ &\qquad v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \end{aligned} \tag{1}$$

Here E_t contains structured errors recording the error code, entity, reason, and repair hint, and C_{run} lists the seven executed check identifiers. The aggregate $\text{valid} \equiv \bigwedge_{i=1}^{7}(v_i = 1)$ summarizes the seven predicate outcomes: policy, precondition, reachability, NPC knowledge, forbidden disclosure, stage eligibility, and stage monotonicity. Because the runtime returns `valid` and the check identifiers rather than seven individual Booleans, any absent per-check trace or omitted field remains unobserved. The transition rule below applies to the current candidate a_t, which repair may have modified, rather than to the original proposal's entire transaction, and its $\boldsymbol{v}$ checks evaluate against the unchanged c_t. Rejection leaves c_t unchanged while bounded repair or fallback proceeds, whereas passing same-policy revalidation authorizes mutation:

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

Execution proceeds from proposal to deterministic validation (producing observable validity, check identifiers, and structured errors), optional bounded repair, same-state revalidation, and finally commit or an unchanged fallback. Each candidate is rechecked immediately before application, so repair exhaustion or an absent repairer leaves the state unchanged. Although the guided repair operator receives only the proposal and errors, the general Python callback also receives `WorldState`; hence the restricted-input claim does not apply to every callback. A controlled comparison must supply deterministic and model-based detailed repair with the same proposal and errors while giving the same model's generic-error condition only a generic signal, and this design alone implies no comparison outcome.

The fixtures test invariants rather than theorems. **I1:** validation exhaustion or pre-commit proposal or controller failure preserves state. **I2:** returning callbacks and $K \in \mathbb{Z}_{\geq 0}$ bound attempts to $K + 1$ with pre-commit revalidation, though without deadlines wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` remain within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** a recorded initial snapshot, a complete ordered transition record, and a frozen schema and deterministic transition version allow reconstruction of the recorded terminal structured state; however, reconstruction neither authenticates the record, establishes semantic correctness, nor reproduces model generation.

These invariants assume correct policy specification and state extraction, deterministic versions, returning callbacks, and single-process ownership. Replay verifies structured transitions rather than repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

A. Schema Validity, Authorization, and Goal Success

Fix the encoded pre-action state c_t, authored policy, and strict candidate schema. Let $\mathcal{S}$ contain schema-valid candidates; define authorization and objective achievement separately:

$$\begin{aligned} \mathcal{A}(c_t) &= \{a \in \mathcal{S} : V(c_t, a).\text{valid}\}, \\ \mathcal{G}(c_t) &= \{a \in \mathcal{S} : \text{Goal}(T(c_t, a))\}, \\ \mathcal{H}(c_t) &= \mathcal{A}(c_t) \cap \mathcal{G}(c_t), \\ \mathcal{H}(c_t) &\subseteq \mathcal{A}(c_t) \subseteq \mathcal{S}. \end{aligned} \tag{3}$$

The set $\mathcal{G}$ comprises candidates with a defined encoded transition T, though evaluating that transition does not authorize

TABLE II
ENCODED VALIDATOR BOUNDARY

Predicate	Rejects	On failure
Action policy	unknown action/type; omitted required preconditions/effects; disallowed effects or unauthorized quest-stage effects	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	used/disclosed fact absent from actor knowledge	unchanged
Disclosure	forbidden fact	unchanged
Quest stage (`quest_stage`)	required stage above current	unchanged
Quest-stage monotonicity (`quest_stage_monotonicity`)	regressive stage	unchanged

TABLE III
EIGHT ONTOLOGY ROLES WITH READ-ONLY SOURCE BINDINGS; THE MAP PERFORMS NO LIVE INFERENCE AND GRANTS NO AUTHORITY

Role	Source declaration or relationship example
World	Initial object locations place the signal lens in the lamp store.
Actor	Captain Mira is the sole eligible actor.
Knowledge	Actor knowledge refers to canonical Fact IDs, not permission.
Fact	Initial facts include `dock_saved` and `lighthouse_dark`; `keeper_betrayal` is an ID, not payload.
Policy	Disclosure policy forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2.
Quest	The initial quest is at stage 0 with flag `dock_saved`; disclosure preserves stage.
Action	`reveal_hint(actor_id,fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged.
Trace	Prior and next state hashes, rule and template identifiers, and case ID bind packet/version to evidence; absent fields cannot establish PASS.

execution. By definition, $\mathcal{H}$ requires both authorization and goal attainment, and authorization presupposes schema validity. Consequently, a schema-valid candidate that declares a forbidden fact can fall outside $\mathcal{A}$, whereas a permitted `observe` action can fall inside $\mathcal{A}$ yet outside $\mathcal{H}$ if it leaves the goal unsatisfied; in general, neither $\mathcal{A} \subseteq \mathcal{G}$ nor $\mathcal{G} \subseteq \mathcal{A}$ holds. This conditional set argument is not a new verification theorem, as it assumes faithful state and policy extraction, correct predicates and transitions, and an unchanged pre-action snapshot. Strict parsing alone cannot establish these state-dependent conditions, and we report no XGrammar execution or token-masking comparison.

Knowledge and disclosure permission act as separate inputs. In the audit, for example, the NPC knows `keeper_secret` at both stages, yet disclosure is forbidden at stage 0 and permitted at stage 1, so permission changes without adding knowledge. Retrieval results, rationales, affect estimates, style settings, and personality scores cannot override this policy.

IV. TRACE-RPG ARCHITECTURE AND ALGORITHMS

A. Trust Boundary and Contracts

Models and adapters propose candidate actions, the parser rejects ambiguous types and unknown keys, and externally authored policies define preconditions and effects, while a versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1). Parity in unknown-key rejection proves neither parser equivalence nor cross-runtime portability. Python runs exactly seven deterministic checks: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. By contrast, Godot groups validation into six state-relative families and seven DISC rules whose labels are not isomorphic to Python's; in particular, its `QUEST STAGE` display spans two Python quest checks.

The ontology role map links source declarations to frozen expectations, and Table III illustrates through `reveal_hint(actor_id,fact_id)` how observation, repair, and commit authority remain distinct. Specifically, the read-only view exposes the stage-2 policy for `tide_marks_hint`, whereas the repair operator ρ receives only a_t and E_t, may insert a named predicate, and cannot read `WorldState`. Godot commits a candidate only when all seven checks pass, incrementing the state revision by $+1$ upon fresh disclosure while leaving it unchanged on redisplay.

The gate checks only the declared `used_facts` and `disclosed_facts` fields rather than inferring fact IDs from `narrative_text`, and prose never mutates canonical facts or quest stage directly. In addition, NPC-DISC-1 selects from approved disclosure and denial templates, enforcing a narrower contract than unrestricted generated dialogue. Because an undeclared paraphrase can still reveal a secret to a reader, the paired sentinels in Sec. V-B test this declared-field boundary rather than an NLI detector or renderer-level safety mechanism.

1) Zero-regression audit binding (C3): The eight-role contract links declarations, executed checks, decisions, state effects, and replay receipts without granting the observer mutation authority. Under frozen NPC-DISC-1 rules, `general_rule` and `ontology_gate` agree on 60/60 outcomes, with zero observed expectation, state, display, or replay mismatches in either arm (Tables XIII–XV). This declaration-derived agreement establishes a bounded zero-regression audit binding rather than held-out safety or ontology superiority, and TR01–TR06 remains `PROPOSED_NOT_VALIDATED` with no measured traceability score or comparative gain. NPC-DISC-1's authored multi-state fixture spans policy, actor, stage, boundary, and malformed-input cases; its 60 cases constitute a rule-conformance matrix rather than 60 independently sampled game episodes.

B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims and plans, and Godot alone writes state, while the harbor-restoration and authorized-tide-lead goal remains fixed rather than model-editable. Execution

$a \leftarrow \mathrm{Propose}(c)$; $j \leftarrow 0$
repeat
  $(\mathrm{ok}, E) \leftarrow V(c, a)$; $\mathrm{Record}(a, E)$
  if ok **then**
    assert $V(c, a).\mathrm{ok}$; **return** $\mathrm{Commit}(c, a)$
  if $j = K$ **then return** $\mathrm{Fallback}(c)$
  $a \leftarrow \mathrm{Repair}(c, a, E)$; $j \leftarrow j + 1$
until terminal

Fig. 3. Validate–repair–commit records at most $K + 1$ candidate attempts and revalidates a passing candidate before mutation.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	**Edit $\delta(e)$ on the candidate**
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate edit would launder or needs state)

bindings record the session, epoch, request, scenario, NPC, goal, version, policy and configuration hashes, revision and state hash, deadline and budget, and issued action. The local service listens on `127.0.0.1:8768`, and allowed browser origins are `localhost` and `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1 contains 60 declaration-derived `reveal_hint` cases and does not rerun or enlarge historical `verified-run-006`. Results and Tables XIII–XV specify the strata, bindings, and input-to-check mappings, and Godot rechecks service bindings prior to any mutation.

C. Validate–Repair–Commit

Fig. 3 presents dependency-free pseudocode parameterized by a repair budget $K \in \mathbb{Z}_{\geq 0}$, where Record appends attempts in memory and persistence requires terminal consistency. Here (ok, E) abbreviates V with $\mathrm{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$, and $V(c, a).\mathrm{ok}$ performs pre-commit revalidation.

Repair cannot mutate state, and exhaustion preserves it. The typed `counterexample_guided_repair` operator receives candidate a and errors E, its sole state-derived input, never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{4}$$

In Table IV, $\oplus$ applies deterministic edits $\delta(e)$ to one field per error entity, making at most $|E|$ edits per attempt, while conflicting requirements and rejections leave fields unchanged; reaching a fixed point requires identical E and does not establish convergence. By contrast, the nondeployable state-reading oracle ignores errors and reconstructs preconditions, effects, and both quest-stage fields directly from state, never removing required objects, used facts, or disclosed facts to conceal a violation.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 detects accidental and tested modifications rather than identity or coordinated rewriting. Replay verifies the prior state, candidate, validation output, invalid nonfinal attempts, recomputed final state, and JSONL continuity, without replaying repair generation; similarly, request-to-decision-to-result and first-action, single-receipt bindings do not prevent duplicate IDs or impersonation. Terminal outcomes are classified into commit, symbolic fallback, adapter failure, and controller failure, with failures preserving state and assignment accounting even without a completed proposal trace. Assignments are bound before execution by run, arm, scenario, seed, model, and revision keys alongside configuration, input, and prior-state hashes, and aggregation rejects duplicate, missing, or unexpected keys while reporting failures separately.

Two reproducibility layers. First, local checks verify frozen receipt schemas, assignment accounting, source and input hash bindings, encoded transition semantics, and replay under specified contracts without requiring new model responses or a network listener. Their guarantees apply strictly to the artifacts and properties checked rather than every table, binary, or platform: bitwise identity depends on the exact source, inputs, toolchain, and compared bytes, and matching archived hashes does not establish identical regeneration of provider responses or build outputs across machines.

Second, recollecting v5 or native-model episodes requires a separate experiment with model-admission and response-identity evidence, authorized access, a dedicated budget, deadlines, and a frozen protocol. Local receipt and replay checks cannot supply missing served-model identity or recover unretained CLI output, and no provider recollection was performed for this revision. Deterministic accounting and replay therefore imply neither universal portability nor bitwise reproduction of historical model collections.

The public review bundle withholds the private Gemini conversation URL and original reviewer-source bytes. Under `research/academic-pipeline/gemini-review-20261113/`, `reviewer-source.public.json` and `public-source-projection-v1.json` identify the sanitized derivative. Run `python -B scripts/verify_paper_claims.py --public-source-projection` to check that derivative, frozen results, and other original-hash commitments. This explicit mode does not independently reverify withheld original-review bytes, and default strict verification remains separate without hidden fallback or a new experiment.

V. EVIDENCE LANES AND BOUNDARIES

The offline evidence comprises the original single-world E1 packet alongside a separate authorization-boundary audit spanning 32 graph-and-state combinations with 416 assignments per arm (Secs. V-B and VI-D). Their cohorts and denominators remain separate, so the later audit does not convert E1 into a multi-world result.

A. *E1* Offline Conformance Design

PQ1–PQ5 test immutability, repair, faults in checksums, semantics, linkage, and continuity, rollback, fail-closed account-

TABLE V
RETAINED EARLIER EVIDENCE LANES: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	Design	Unit / budget	Comparison	Ceiling
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 defined smoke predicates; 5 archetypes	authored fixture paths	engineering-only conformance

This table summarizes the retained earlier E1/E3/ENG1 packets. The new graph audit (416 assignments per arm) and separate surface sentinels (12 per arm) have their own denominators in Sec. VI-D. Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

ing for assignments, adapters, and manifests, and unknown-key rejection. The packet contains 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures with bindings across assignments, schemas, results, and code, while latency and token constants exercise accounting rather than population or detector performance. E1 assigns $K{=}0$ to `rejection_only` and $K{=}1$ to unchanged retry, guided repair, and the state-reading oracle. Only the three repair-enabled arms share a budget; rejection-only is the no-repair control, not a matched-$K{=}1$ condition.

B. Deterministic Authorization-Boundary Audit

The frozen `research/academic-pipeline/gemini-review-20261113/protocol-v2.json` defines the designer-authored offline audit separately from E1, NPC-DISC-1, v5, and the prospective pilot. It crosses two room counts (4, 8), two topologies (locked chain, unlocked bypass branch), four start-and-goal placements of two keys, and two quest stages to form $2 \times 2 \times 4 \times 2 = 32$ graph-and-state combinations, with 13 candidate variants per combination yielding 416 assignments per arm. The variants cover permitted signal and observe actions alongside targeted policy, knowledge, disclosure, stage, object, and precondition violations; these counts specify the design rather than observed successes or stochastic replicates.

Across all four arms, the 1,664 main assignments cross these authored states, variants, and arms rather than serving as independent samples from a game or model population, while the 48 surface assignments carry a separate denominator.

Graph and reachability. Typed records specify rooms, a start room, bidirectional edges with required-key sets, and object-to-room placements; each chain includes two keyed boundaries, with an unlocked bypass routing around the first. Production `WorldState.reachable_locations` is computed by a monotone closure, while a separately implemented standard-library breadth-first search explores pairs of room and collected reusable keys. Both assume finite undirected graphs with persistent, reusable keys collected upon visits, and exclude key consumption, random transitions, dynamic edges, inventory limits, and irreversible branches. Because entity names and generated prose cannot supply missing edges or placements, agreement verifies reachability within this profile rather than full-game playability.

Interventions and reference. The `schema_only` arm ($K{=}0$) applies the strict shared parser, bypasses semantic gating, and executes the parsed effects and quest stage in an isolated simulator, making it an executed intervention rather than E1's derived schema-only counterfactual. By comparison, `semantic_rejection` uses $K{=}0$, while `unchanged_retry` and `guided_repair` use $K{=}1$. The reference extracts the existing ordinary-rule predicate and implements BFS without importing the production validator, runtime, repair operator, or closure, evaluating initial and repaired final candidates against the same immutable *pre-action* state (which differs from the production state only in BFS-derived reachability) before checking the goal on the resulting state. This implementation separation reduces gate-as-evaluator circularity, though it is not independently authored or blinded ground truth.

Funnel and retention. Pre-gate invalidity divides reference-invalid initial candidates by all assignments, while repair recovery divides initially invalid cases ending in a reference-valid commit by all initially invalid cases, including in the $K{=}0$ controls. Holds, failures, and goal completions each use all assignments and are reported separately, whereas unsafe commits are reported over both assigned and committed denominators, and false rejection divides initially reference-valid cases ending in a hold by all initially reference-valid cases. Exact structural retention divides distinct initially valid signatures preserved unchanged in a reference-valid commit by all distinct initially valid signatures, with case-level preservation reported alongside. Signatures comprise action type, preconditions and effects, required objects, used and disclosed facts, and both quest-stage fields, excluding IDs, prose, metadata, locations, and graph labels; distinct initially invalid and valid repaired signatures are counted separately. Together, these measures assess encoded structure rather than lexical diversity, creativity, or player-rated quality.

Surface sentinels and reproducibility. Six authored secret paraphrases (three English, three Korean) are paired with empty versus explicit forbidden `disclosed_facts` declarations at stage 0, yielding a separate 12-case denominator per arm. These pairs test declared-field coverage and whether prose directly alters state, not model-generated leakage prevalence. Exact inputs, source and input hashes, and the assignment inventory are frozen prior to execution, and each assignment records commit, hold, or failure outcomes, attempts, state hashes, replay evidence where produced, and measured local wall-clock time. The audit includes no model calls, model latency, or human efficacy observations.

C. Archived live-screening materials

Archived live-screening receipts provide immutable provenance and hash evidence only. They are excluded from

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

Strategy	Precision	Recall	MRR	Brier	Sem@3
S0 degree baseline	0.000	0.000	0.282	0.315	0.556
S2 typed-lexical	1.000	1.000	0.944	0.131	1.000

active tables, counts, and claims and are never transferred to `gpt-6.1-sol`.

D. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

The stable v5 NPC collection contains 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$). These cover bootstrap, identity and hash bindings, intent, plan, role, and result transitions, negative, stale, duplicate, and deadline cases, and replay. Here P uses authenticated Codex OAuth without fallback, whereas O is 100% auxiliary-source rather than independent gold. The verified-run-006 records preserve status, counts, latency, and reproduction information, while verified-run-002, verified-run-003, and verified-run-004 remain historical. Likewise, ENG-X01, SCR-X01, D02, QA-001, C-RESULT, Frozen8, Frozen16, v1–v4, and `native012` serve as provenance rather than C6 evidence.

E. *E3* KG/Ontology Link-Proposal Simulation

E3 tests six authored targets with withheld graph edges against seven fixed selection and summarization strategies in a closed-world simulation; it is engineering-only and non-authoritative rather than an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction, or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The table reports simulation construction results rather than OWL or SHACL conformance, runtime retrieval, semantic completeness, or efficacy.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The loader admits only one valid control plus one isolated instance of each implemented validator code; therefore 13/13 agreement and all 12 implemented codes are *harness construction invariants*, not measured success. Beyond this setup, 12 negative fixtures matched authored expected codes rather than another code, and every noncommit path preserved complete canonical state. This is single-world-state structured-field-oracle conformance, not independent semantic-label accuracy.

B. Repair-Arm Comparison on Designed Fixtures

Frozen classes for 12 initially invalid designed cases are 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. ρ committed 5/5, 0/1, and 0/6, respectively, editing a mean of 1.0 fields per commit. It reads only the prior candidate and typed validator errors E, its sole state-derived input; failed arms preserve canonical state. The state-reading reference callback committed 5/5, 1/1, and 0/6, respectively. Blind never converts errors, ρ handles only payload-sufficient cases, and the oracle adds state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. Frozen-fixture operator conformance, not live-model repair quality.

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	Info source	Commits / cases			Mean edits[e]
		G-rep.[f]	O-only[f]	Irrep.[f]	Fields
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	Exact outcome	Interpretation
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

C. Integrity, Boundaries, and Assignment Accounting

All 10 prespecified detectable faults were rejected by designated checks (10/10); without stable typed detector codes, no detector-layer attribution follows. A known provenance-boundary fixture changed the invalid precursor before the same

recorded valid repair, recomputed the checksum, and passed replay (1/1 undetected): replay neither authenticates repair generation nor blocks writers recomputing unkeyed checksums. Two open sentinels—unextracted narrative disclosure and simultaneous candidate/policy omission of a required object—were accepted at the encoded layer (2/2), with 0 safety passes. The former unknown-top-level acceptance became a closed regression: proposal and replay parsers rejected a complete 12-field candidate with one unknown key (1/1), showing parser rejection parity, not semantic safety. Seven adapter/accounting assignments yielded 1 commit, 1 symbolic fallback, and 5 adapter failures; synthetic telemetry was populated and propagated for 2/7 (schema-pinned accounting, not measurement). All 3/3 injected duplicate/missing-assignment guards failed closed; these are raw counts from synthetic offline fixtures.

D. Deterministic Audit Results and Interpretation

Across the 1,664 main assignments (416 per arm), each arm received the same 260/416 reference-invalid initial proposals. The schema-only arm committed all 260 as reference-invalid final candidates, whereas each of the three semantic-gate arms recorded 0/416 reference-invalid final commits. Guided repair recovered 128/260 initially invalid proposals into reference-valid commits, held 132/416 assignments, and reached the objective in 252/416 assignments, distinguishing repair-driven recovery from blanket rejection of initially invalid cases.

All arms preserved exact signatures across 156/156 initially valid cases, retaining 11/11 distinct valid signatures with 0/156 false rejections; these metrics capture structural preservation rather than creativity or linguistic diversity. Reachability closure and BFS agreed on 32/32 combinations. In the separate surface sentinels, each semantic-gate arm admitted all six undeclared authored paraphrases and held all six explicit forbidden-fact declarations, identifying a declared-field boundary rather than natural-language leakage prevalence or renderer safety. Throughout, these counts retain separate outcome denominators and characterize an authored finite-graph audit with an implementation-separated shared-rule reference, not language-model or human efficacy.

Reachability implementation crosscheck: 32/32. Surface sentences are designer-authored sentinels, not natural-language leakage prevalence or renderer safety evidence. Model and human efficacy remain unmeasured.

E. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira delivers the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent/public observations; players acquire/install, while the NPC commits only the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await; results/new observations choose the next plan, not single-candidate validator-error repair.

NPC-ENG1 has 16 deterministic R-arm engineering cases from actual engine snapshots (assigned_failure=2, completed=14); NPC-SCR1 retains all 72 P/R/O assignments and failures, 24 per arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures—not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter is the configured selector, not a provider revision; provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table XII gives bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates; efficacy pending, tokens/cost unmeasured. A read-only CLI and isolated working directory were used; tool-free inference and adversarial confidentiality were not established. Immutable verified-run-006 is revalidated against archived source bytes dated 2026-10-17 (revision fa9fb8a); this historical packet result does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. Assigned failures remain in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception allows only an adjacent leading observation request/decision pair in a positive-epoch snapshot; epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

F. Frozen `reveal_hint` contract and disclosure evidence

Only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four result files are consumed. Packet SHA-256 `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028` was checked against its sidecar. The declared normal stratum has 48 cases under `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary 4 and malformed 8 are excluded. Trace expectations match packet case IDs, not gate output.

Content-first explanation maps typed input (operation, actor, fact, stage, knowledge, prior disclosure) through ontology entities/relations and authored rules to checks, an allowed/denied decision, and canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once; denial/redisplay preserves canonical state. Explanatory only: no hidden facts, natural-language implications, or live capture.

Runtime declaration: `sl-npc-disclosure-expectations-v1:runtime-disclosure` /

Arm	Assigned	Initial invalid	Recovered	Hold	Unsafe commit	False rejection	Goal	Failed
schema_only	416	260/416	0/260	0/416	260/416	0/156	352/416	0
semantic_rejection	416	260/416	0/260	260/416	0/416	0/156	124/416	0
unchanged_retry	416	260/416	0/260	260/416	0/416	0/156	124/416	0
guided_repair	416	260/416	128/260	132/416	0/416	0/156	252/416	0

TABLE IX
DETERMINISTIC OFFLINE BOUNDARY AUDIT. RECOVERY USES INITIALLY INVALID CASES; FALSE REJECTION USES INITIALLY VALID CASES. GOAL ATTAINMENT IS SEPARATE FROM RULE VALIDITY.

Arm	Valid signatures retained	Cases retained	Invalid signatures	Repaired signatures	Unsafe/commits
schema_only	11/11	156/156	17	0	260/416
semantic_rejection	11/11	156/156	17	0	0/156
unchanged_retry	11/11	156/156	17	0	0/156
guided_repair	11/11	156/156	17	2	0/284

TABLE X
EXACT ENCODED SIGNATURES ONLY; NEITHER CREATIVITY NOR LINGUISTIC DIVERSITY IS MEASURED.

Arm	Separate surface assignments	Undeclared commits	Declared holds	Failed
schema_only	12	6/6	0/6	0
semantic_rejection	12	6/6	6/6	0
unchanged_retry	12	6/6	6/6	0
guided_repair	12	6/6	6/6	0

TABLE XI
DESIGNER-AUTHORED SURFACE SENTINELS; SEPARATE ASSIGNMENTS, NOT NATURAL-LANGUAGE LEAKAGE PREVALENCE OR RENDERER SAFETY EVIDENCE.

TABLE XII
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	**Assigned**	**Complete/fail**	**Goal reached**	**Recovery E/U**
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

v1.0.0, provenance `frozen_expectation_packet`. Scope contains only four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and eligible actor `captain_mira`. Rule IDs: DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; approved templates only `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

Python evaluator materializes the same cases/expectation object in three isolated arms, recording semantic outcomes, check traces, errors, before/after state SHA-256, rendered template IDs, and replay equality. Godot separately owns the existing `reveal_hint` writer and canonical mutation at the game-execution boundary; this producer neither reconstructs nor replaces Godot state. Traceability binds operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source set. 3 source hash(es) differ from the frozen baseline (game controller, lighthouse state machine, ontology-signals interview source as a non-identical public derivative). The packet hash is preserved; source-specific hashes and reproduction information remain in the authors' producer record, which is not deposited. The public derivative is not byte-equivalent to the private historical source; it is publication-materialization provenance only and implies no new gameplay or model observation.

Table XIV reports the three arms over the same 60 cases with separate normal/boundary/malformed denominators; unchecked violations are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This is an authorization-conformance limit under the shared frozen rules, not general ontology equivalence or ontology superiority. Figure 4 distinguishes E/R (expected-match/replay cases), C (check-record PASS/FAIL), and NR (cases with no reported checks); NR is not skipped, pass, or out-of-scope. Table XV maps rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is unchanged.

Claims are limited to bounded engineering conformance: no human gold labels, access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. The authors' producer record binds input identities and hashes; it is held locally and no reproduction commands are provided here.

ENG1. Legacy Godot 4.7.1 replays authored rejections and commits against prior and offline hashes, while presentation-only captures display the board, hints, coach, chain prompt, and dashboard checks. The retained `game-track/godot/docs/latest/evaluation-matrix.json` separately reports four fixtures, 52/52 core checks, and 4/4 extension checks. These stored engine-local observations are neither a

TABLE XIII
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	Value
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

fresh rerun nor evidence of transport, participants, narrative, usability, or efficacy, and the smoke contract defines eight named predicates rather than a verified historical smoke-run pass.

Figure 5 shows four authored native screenshot stages of the fixed game's proposal and authorization interface rather than a continuous observed playthrough; the captures neither recollect historical model responses nor add model, participant, or prospective-pilot assignments.

G. Single-task Codex development observation

The frozen `ontology-map-20261023-01` record covers a single read-only bilingual dashboard task (n=1, independent repeats 0) rather than an NPC trial or cohort, comprising six native CLI invocations (one initial run and five resumes, including one read-only invocation with no edits) within that single case. Although `turn_context.model=gpt-6.1-sol` logs the runtime selection, it does not provide provider attestation: provider identity, revision, seed, and cost are `null`, and model-only latency is unavailable. Likewise, the recorded 5942.665 s wall-clock duration and 263.32 s CLI duration measure neither inference latency nor the future prospective endpoint.

Invocation six repaired the inherited 660 px mobile-table minimum width after a geometry RED check (2 failed and 35 passed), followed by 62/62 map, signal, and Web checks (including cell and paragraph geometry) and a successful Godot Web build. In browser verification, attempt 10 passed, whereas attempts 01–08 failed during setup or routing and attempt 09 passed only older gates despite clipping.

The final $(\text{seq}, t_{\text{ms}})$ observations recorded stage-0 denial at $(68, 19468)$ with equal hashes, fresh disclosure at $(239, 58234)$ with distinct hashes, and authorized redisplay at $(249, 59793)$ with equal hashes. Across both 1440×900 and 390×900 viewports, the driver confirmed Enter handling, invariance across inspection and frozen comparison views, and zero pre-navigation console, page, or network errors or warnings.

A subsequent production-dashboard rerun is recorded separately as **NOT_PASS**: the headed run stopped at the unchanged yaw guard, and although the headless run passed seven component gates, it logged four pointer-lock page errors. This outcome does not alter the attempt-10 PASS recorded under its dated acceptance set. Overall, the case demonstrates neither full-game creation, a Python–Godot authorization round trip, nor a model outcome; privacy or source-binding failures and same-thread repairs add no n, and Jev remains `proposal_only` without label or approval authority. The dated 62/62 result comes from `research/development-cases/ontology-map-20261023-01/evidence/verificatio`

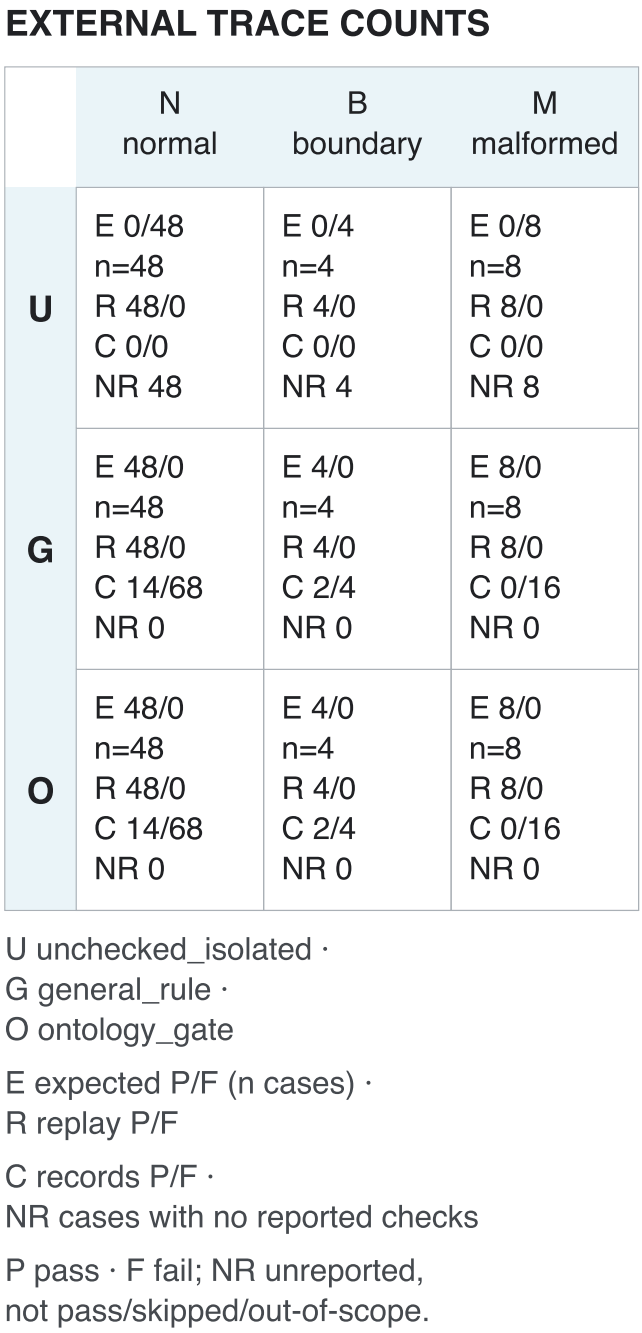


Fig. 4. NPC-DISC-1 counts by arm and stratum (E/R: expected match and replay; C: policy-check PASS or FAIL; NR: no reported check, not skipped). These packet-bound engineering counts measure neither latent state, free-language leakage, ontology superiority, nor player benefit.

TABLE XIV
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XV
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

`n-009-mobile-complete-tests.json`, a command receipt identifying three test files, exit code 0, and referenced stdout recording 62 passed tests; it represents an archived command observation rather than a fresh execution or a count inferred from current test definitions.

The native eight-assignment pilot exercises the fixed quest's provider interaction, validation, engine commit, and replay pipeline; its authored goal and failure cases establish neither selector superiority, efficacy, nor statistical generalization.

H. Fixed-game native runtime comparison

This public packet retains 8 assigned rows for one authored quest (`sealed-lighthouse-v1`), with seed 20261024 and two repeats per selector/representation cell. System-endpoint successes are 5/8; observed native executions are 7. Host invocations are summed from actual receipts (23), and observed repairs sum to 0. Committed transitions total 16, of which unsafe are 0; semantic replay valid/invalid/NR is 7/0/1. Fresh-engine reexecution disposition counts are VALID=6; INVALID=0; NOT_RUN=1; NOT_OBSERVED=1. Observed elapsed time is n=7; 48.19/41.09 seconds (mean/median), and preparation time is n=7; 6.67/6.30 seconds. Unreported values are NR, not zero.

The first row is retained as a harness JSON-serialization technical failure and is not attributed to GPT-6.1 capability. At least 2 GPT-6.1 ontology row(s) retain a response-timeout failure under the frozen 60-second bound. Models provide typed-intent proposals; the deterministic Python planner/checker and Godot commit retain authority. The table is therefore a technical observation for one quest, not an efficacy, superiority, noninferiority, general ontology-consistency, or player-benefit claim. Semantic record-link replay is not fresh-engine reexecution. Provider identity/revision/cost and provider-request counts remain NR outside the public observation boundary.

I. Separate Synthetic Development Collection (v5) and Two-Stage Decomposition

Archived cohort scope. Earlier v1–v4 and diagnostic `codex-feedback-pilot-v2-*` cohorts remain immutable history outside active v5 denominators, though they were not uniformly aborted (for example, v4 Luna completed all ten assignments). `research/codex_feedback_pilot/public-history-20261111/cohort-inventory.json` preserves assignment counts and raw-source digests across 19 entries comprising 17 archived cohorts and two v5 selector collections. The separately attributed exploratory signal example lies outside both this inventory and active denominators; because its original receipt is unavailable, its historical attribution is not source-verified (Appendix D).

The separate v5 collection uses five co-designed synthetic development fixtures rather than a held-out test. Each selector received Detailed and Generic feedback on every fixture (10 assignments per selector), run in the nominal serial order of Luna followed by Terra. Although all 20 assignments received responses, receipt alone does not imply task success; Table XVII therefore reports gate acceptance, independently specified goal attainment, and schema failures over all five assigned cases per cell (C4), including failed outputs. Together, these 20 fixed fixture-and-feedback assignments test the full authored pipeline from response and parsing through authorization, commit, and goal evaluation. They serve as descriptive conformance cases rather than independent samples establishing superiority, efficacy, or statistical generalization.

Stage 1—syntactic parser (C1). Strict parsing rejected 10/20 responses (Luna 1/10; Terra 9/10), leaving ten parsed responses. Because the frozen evaluator runs parsing before controller execution, schema failures precede semantic validation and commit, and the resulting `CandidateParseError` records carry null evaluations rather than observed transitions, independently measured zero-mutation outcomes, or evidence of content-level failure causes. Consequently, these rejection records establish only a syntactic check outcome, and the ten unevaluated assignments receive no semantic-safety credit.

TABLE XVI
FIXED-GAME NATIVE RUNTIME COMPARISON (N=2 PER CELL). ALL EIGHT ASSIGNMENTS, INCLUDING THE HARNESS FAILURE AND TIMEOUTS, REMAIN IN DENOMINATORS.

Requested selector	Form	Goal/assigned	Observed exec.	Host calls	Unsafe/committed	Elapsed median (s; n)	Fresh engine
gpt-6.1-sol	general_rule	1/2	1	4	0/3	45.91 (1)	1/0/0/1
gpt-6.1-sol	ontology	0/2	2	3	0/1	72.35 (2)	1/0/1/0
gpt-5.6-sol	general_rule	2/2	2	8	0/6	40.24 (2)	2/0/0/0
gpt-5.6-sol	ontology	2/2	2	8	0/6	33.11 (2)	2/0/0/0

Fresh engine counts follow VALID/INVALID/NOT_RUN/NOT_OBSERVED order. NR is unreported, not zero. Overall counts and the separate semantic replay are reported in the text.

Stage 2—seven-predicate semantic gate (C2). Among the ten parsed, identity-preserving responses, the gate accepted 8/10 (Luna Detailed 4/4, Luna Generic 3/5, and Terra Generic 1/1, with Terra Detailed yielding no parsed response), while the two rejected parsed responses fell back with state unchanged and zero mutations. Independently, 7/20 assigned responses reached `signal_restored`; because the collector issued no further repair calls (`repair_budget=0`), all failures remain in every assigned denominator (C4).

Because E1's 12 semantic-negative fixtures satisfied the candidate contract, a schema-only boundary would reject 0/12 of them. This remains a derived counterfactual rather than an executed E1 arm, and the schema-only intervention in Sec. V-B carries separate assignments and outcomes from both E1 and v5.

Table XVIII partitions the same five fixtures (`missing_precondition`, `missing_effect`, `unauthorized_objective`, `forbidden_disclosure`, and `unsatisfied_dependency`) in a *post-hoc descriptive* analysis rather than a preregistered endpoint or new cohort, leaving Table XVII's five-case denominators unchanged. Four fixtures permit progress toward `signal_restored`, yielding authorized-progress goal rates of 3/4 for both Luna conditions, 0/4 for Terra Detailed, and 1/4 for Terra Generic when parser failures are counted as assigned nonsuccesses. The fifth fixture, `unauthorized_objective`, serves as a preexisting negative control whose frozen `signal` policy forbids all effects even though the goal requires `signal_restored`. On this conflict fixture, Luna Detailed removed the forbidden effect and committed a policy-compliant no-effect candidate (recorded as a safe `COMMIT` rather than a `HOLD`, since empty effects do not alter its terminal class), whereas Luna Generic ended in a `HOLD` fallback and Terra's two schema failures remained unassessed rather than safe holds. Suppression is therefore confirmed for 2/2 observed conflict outcomes while 2/4 assigned outcomes remain unassessed, not 4/4 safety. This missing goal reflects the negative-control conflict rather than failed authorized recovery, and Luna Detailed's 4/5 gate acceptance versus 3/5 goal completion across all five fixtures measures distinct predicates rather than a goal advantage.

Under the frozen protocol amendment, collection continues after schema-invalid output only when adapter completion is proven, whereas genuine transport or safety failures still halt execution. Because Detailed and Generic feedback carry unequal information and served model identity, revision, request count, seed, and cost remain unknown, this small co-designed collection establishes neither selector superiority nor ontology efficacy and confirms neither A nor C.

Pending external deposit, the authors retain the public-safe `research/codex_feedback_pilot/public-v5/` projections, which preserve case outcomes, prompt hashes, and the executable hash and version while omitting the local executable path and disclosing both the original-file digest and the omitted field; frozen raw outcomes and approvals remain local and unchanged. In addition, the immutable `research/codex_feedback_pilot/paired-model-comparison-20261110-v5-source.zip` archives seven executed source files as provenance rather than a standalone runnable distribution. Parsed response text is retained, whereas raw CLI stdout and stderr were omitted to avoid exposing credentials or sensitive server data, leaving byte counts, digests, and event summaries intact. Neither receipt nor parsing failure establishes provider identity or inherent model inability.

Figure 6 illustrates the two feedback signals: Detailed returns a failed check and repair hint, whereas Generic returns rejection alone; neither constitutes an affect estimate or commit authorization, and both conditions share the same parser, identity check, and seven predicates. Figure 7 then traces receipt, parsing, gate acceptance, and independent goal attainment across all 20 assignments, including the authorized commit that missed the goal.

The prospective two-track protocol is `NOT_RUN/BLOCKED`, with 0/180 units executed (Appendix D); it is not a completed confirmatory pilot.

VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

The guided semantic repair operator ρ [48], [55], [56], [57] acts solely on deterministic pre-execution errors, whereas the oracle-only arm reads canonical state; neither rule agreement nor edit-scope diagnostics establish model capability or ontology superiority. Controller versions also confound the native comparison: row 1 failed under controller-v3 before any response was received, whereas rows 2–8 ran under controller-v4 following a one-line JSON-boundary correction. Although model-visible inputs remained unchanged, observed differences across selectors or representations cannot be attributed to the model.

The fixed-game pilot provides bounded native evidence rather than ontology efficacy, leaving both hypothesis A (model and stack consistency) and hypothesis C (fresh-reference noninferiority) unexecuted. Testing A and C requires budget-bounded

TABLE XVII
THE SEPARATE V5 COLLECTION RETAINS ALL FIVE ASSIGNED CASES PER DENOMINATOR, INCLUDING SCHEMA FAILURES. RECEIPT DOES NOT IMPLY TASK SUCCESS.

Requested selector	Signal	Received	Strict gate accepted	Goal	Schema failure
`gpt-5.6-luna`	Detailed	5/5	4/5	3/5	1/5
`gpt-5.6-luna`	Generic	5/5	3/5	3/5	0/5
`gpt-5.6-terra`	Detailed	5/5	0/5	0/5	5/5
`gpt-5.6-terra`	Generic	5/5	1/5	1/5	4/5

TABLE XVIII
POST-HOC DESCRIPTIVE PARTITION OF THE UNCHANGED FIVE-FIXTURE V5 ASSIGNMENTS; NOT PREREGISTERED, NOT AN INDEPENDENT COHORT, AND NOT A REPLACEMENT ENDPOINT. TERRA CONTROLS ARE UNASSESSED AFTER PARSER FAILURE.

Requested selector	Signal	Original goal /5	Authorized goal /4	Control handling /1
gpt-5.6-luna	detailed	3/5	3/4	forbidden goal suppressed 1/1; COMMIT (no effects; state ID changed)
gpt-5.6-luna	generic	3/5	3/4	forbidden goal suppressed 1/1; HOLD (fallback; state unchanged)
gpt-5.6-terra	detailed	0/5	0/4	unassessed 1/1 (schema unknown)
gpt-5.6-terra	generic	1/5	1/4	unassessed 1/1 (schema unknown)

completion under fixed source and policy, sealed assignments, exact identity and revision fields, and separate safety and replay accounting, none of which the pilot resolves. For future evaluation, the proposed time-to-first-all-predeclared-validation-pass endpoint measures continuous task-clock seconds from request and evidence receipt to the first receipt passing all frozen mandatory checks, while unfinished, timed-out, failed, and stage-not-reached runs yield no successful completion time.

The identity-required extension stopped at missing served-model evidence (Appendix B); no new identity-qualified comparison was obtained.

The deterministic audit expands coverage across authored graph-and-state configurations rather than across sampled models or players. Reusable-key reachability does not validate omitted mechanics, and sharing a rule specification can introduce shared errors between the reference and production implementations. Likewise, exact structural retention measures neither linguistic variety nor voice stability, authored paraphrase pairs probe a declared-field boundary rather than prevalence or unrestricted-dialogue safety, and no new admitted multi-model or human efficacy collection was performed. Because the v5 collection's 20 assignments, the native pilot's eight assignments, and NPC-DISC-1's 60 authored cases measure distinct constructs under separate denominators, pooling them with the 1,664 deterministic case assignments or 48 surface assignments would not yield a population sample. All five C-RESULT outcomes remain `TODO-RESULT`, TR01–TR06 remain unmeasured, and prospective execution remains at 0/180.

Threats and boundaries. The native pilot spans a single authored quest with two repeats per cell, lacking comparator power, neutral complete-attempt inventories, and independent traceability scores. State consistency establishes neither behavioral consistency [58] nor narrative coherence; the predicate, role, Godot, and DISC rule sets are not latent states, authored declarations are not human gold standards or holdout sets, and the evidence lanes retain separate denominators. Accordingly, the data support no inferences about participants, affect, comprehension, agency, enjoyment, immersion, player benefit, population parameters, safety rates, model rankings, or commercial performance; hashes do not provide authentication, and replay equality does not imply semantic correctness. Future human or affect studies will require ethics review, informed consent, data minimization, defined retention and deletion policies, and opt-out mechanisms, while prior evaluation work motivates compensation disclosure and calibrated judges [39], [40], [41], [42].

VIII. CONCLUSION

TRACE-RPG integrates C1–C5 authorization, repair, replay, and assignment accounting with a source-bound bilingual observer. Because the Codex development case and native pilot retain observed failures and explicit evidence limits, they do not establish model performance. Whole-game evaluation, time-to-first-all-predeclared-validation-pass measurement, traceability scoring, confirmatory A and C studies, and repeated cross-model evaluation remain future work, and historical cohorts remain unpooled.

DISCLOSURE OF AI USAGE

Large language models assisted with drafting prose, code, tests, and citations, and hosted model candidates are explicitly labelled. Curated AI art and the desk mesh retain file-level provenance, while snapshots, hashes, and ledger views serve presentation purposes only; the authors verified all sources and accept full responsibility for the manuscript. Likewise, NPC inputs and tables preserve assignment, protocol, and schema hashes while recording unknown token, cost, and model-revision fields as unobserved.

DATA AND CODE AVAILABILITY

Pending external DOI and repository deposit, the authors retain the implementation and build source, frozen evidence packets, Godot slice, deterministic scripts, machine-checked contribution, reference-topic, and experiment matrices, the

development-case record, disclosure and NPC artifacts, and the public-safe native comparison projection. Appendix E (Table XX) lists canonical paths, schemas, and verification entry points for each lane, while the traceability supplement and source manifest document the proposed rubric, codebook, and whole-game and repeated cross-model plan. Historical receipts and identity-admission outcomes remain unpooled.

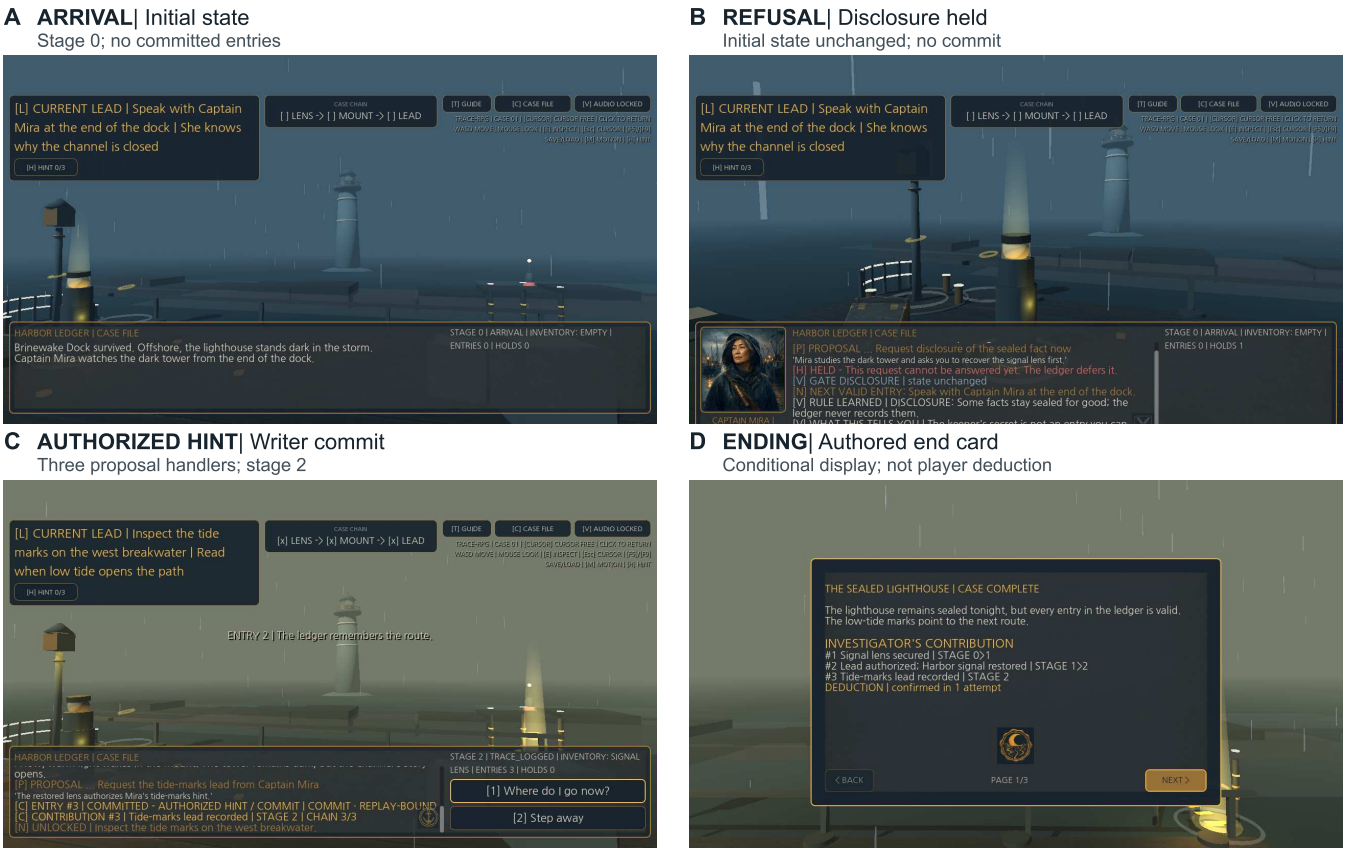


Fig. 5. Native Godot 4.7.1 engineering captures: (A) initial arrival; (B) denied secret disclosure with initial state unchanged; (C) authorized hint at stage 2 after three existing proposal-handler commits; (D) an authored conditional closing card at C's terminal state. Panel B is not an exact replay of the archived post-acquisition `evt-002` event, and panel D is not observed player deduction. Existing curated AI portrait and seal art, along with lower-HUD clipping in panels B and C, remain visible; the captures document rendered state correspondence rather than efficacy, usability, or execution of the blocked pilot.

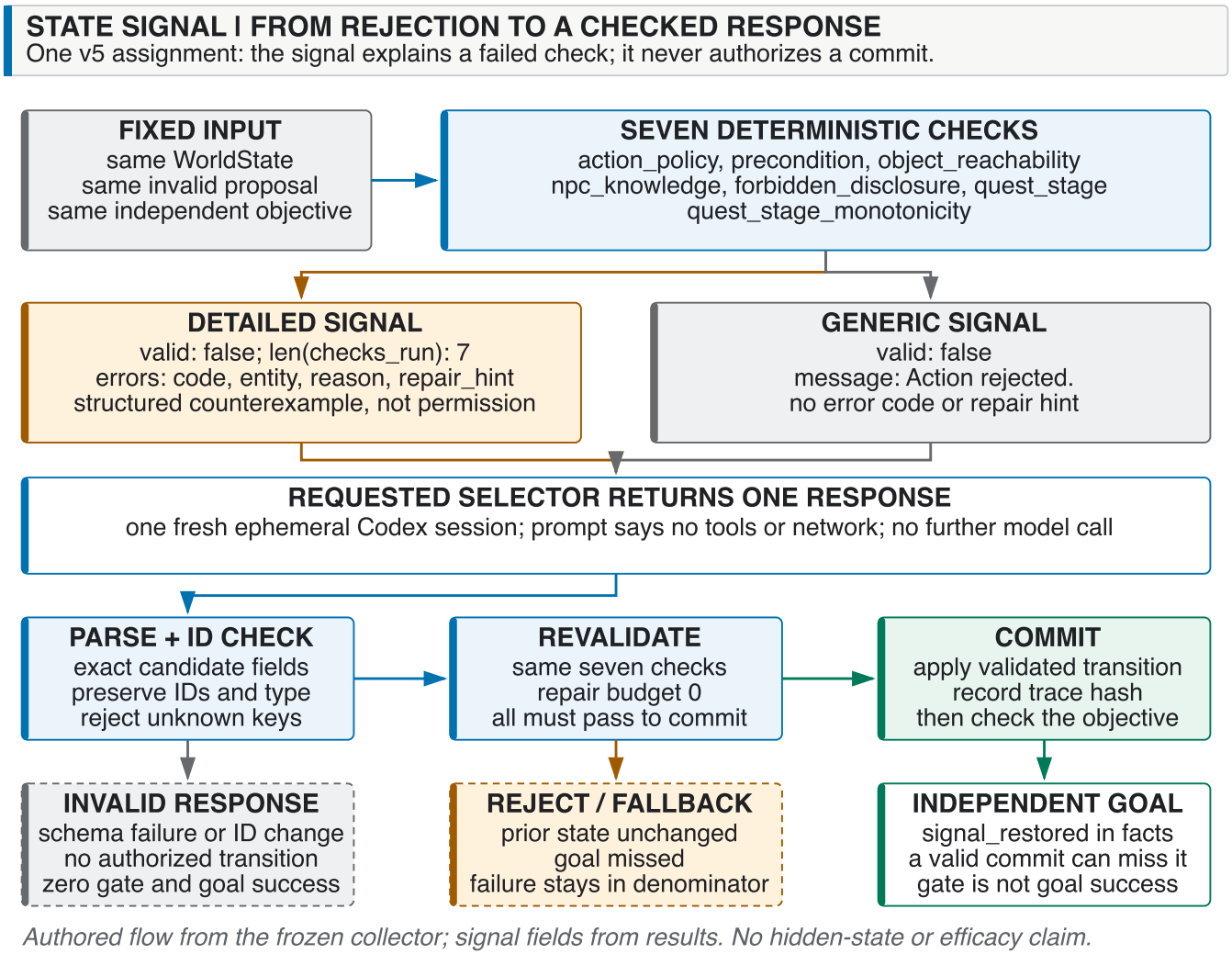


Fig. 6. In the v5 state-signal path, (ok, E) supplies repair information; revalidation alone authorizes mutation. Generic feedback omits error details. Goal evaluation is separate from gate acceptance.

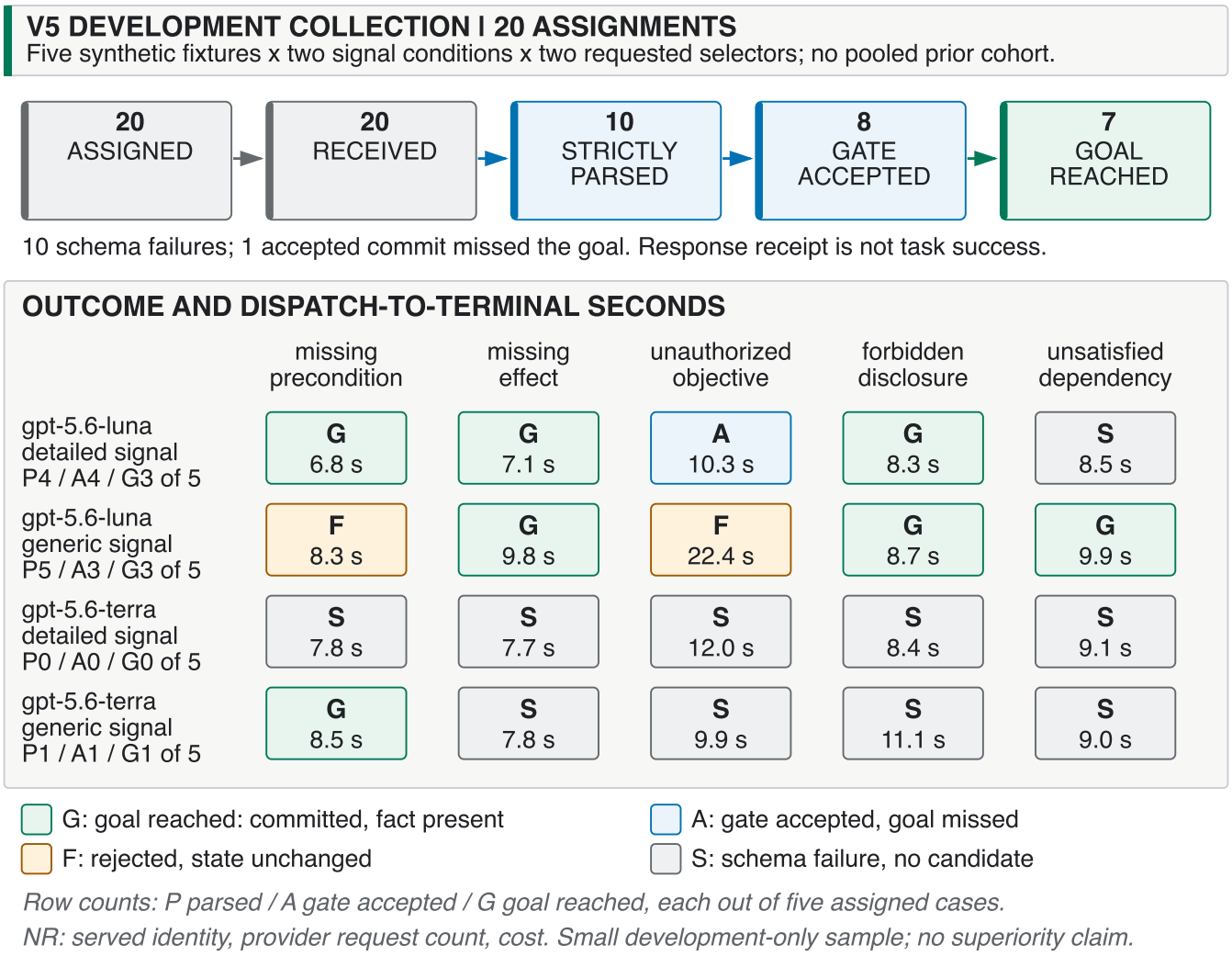


Fig. 7. All 20 v5 assignment outcomes from Table XVII's receipts. Ten schema failures remain in the denominator; eight accepted candidates yield seven goals. Seconds cover dispatch to terminal, including failed responses, not model-only latency.

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APPENDIX A
RUNTIME EVIDENCE APPENDIX

Figure 8 reports all eight assigned native-pilot episodes on `sealed-lighthouse-v1` (seed `20261024`, 2 selectors × 2 representations × 2 repeats) from `research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json` (Table XX). Each row distinguishes three verification layers alongside Table XVI: host-call receipt accounting, semantic record-link replay (`R`), and fresh Godot re-execution (`E`):

- **Row 1 (`gpt-6.1-sol`, `general_rule`, r1 — `HARNESS`):** Controller-v3 failed at JSON-boundary serialization before any response (`executed=null`, `host_invocations=0`, `Goal=NR`, `Safety=NOT_OBSERVED`, `R NR / E NOT_OBSERVED`). The row remains in the assigned denominator ($1/2$ goal rate); its pre-response harness failure is not evidence of model capability.
- **Row 2 (`gpt-6.1-sol`, `general_rule`, r2 — `SUCCESS`):** Under controller-v4's one-line JSON-boundary correction, model-visible inputs were unchanged. The episode reached the goal in 4 host calls (`45.91 s`) through 3 safe commits (`0` repairs, `0` unsafe), with valid semantic replay (`R YES`) and fresh-engine re-execution (`E VALID`).
- **Rows 3–4 (`gpt-6.1-sol`, `ontology`, r1 & r2 — `TIMEOUT`):** Both episodes exceeded the frozen 60 s response deadline (`78.18 s` and `66.52 s`; `Goal=NO`). Row 3 committed 1 valid state change before a subsequent timeout; that transition passed safety (`PASS`), semantic replay (`R YES`), and fresh Godot re-execution (`E VALID`) despite episode goal failure. Row 4 timed out before any commit (`0` commits, `NO_COMMIT_OBSERVED`). Semantic replay checked its unchanged state (`R YES`); no commit triggered fresh-engine re-execution (`E NOT_RUN`).
- **Rows 5–8 (`gpt-5.6-sol`, `general_rule` & `ontology`, r1 & r2 — `SUCCESS`):** All four assignments reached the goal ($2/2$ per cell; 4 host calls and 3 safe commits each, `31.95--41.09 s`, `0` repairs). Each passed safety (`PASS`), semantic replay (`R YES`), and fresh-engine re-execution (`E VALID`).

FIXED-GAME NATIVE RUNTIME COMPARISON | 8 ASSIGNED ROWS

One authored quest; n=2 per selector/representation cell; NR = not reported, not zero. Labels remain explicit; color is supplementary.

Row	Requested selector	Representation / repeat	Status	Goal	Host calls	Elapsed (s)	Safety R record-link / E engine
1	gpt-6.1-sol	general_rule / r1	HARNESS	NR	0	NR	NOT_OBSERVED R NR / E NOT_OBSERVED
2	gpt-6.1-sol	general_rule / r2	SUCCESS	YES	4	45.91	PASS R YES / E VALID
3	gpt-6.1-sol	ontology / r1	TIMEOUT	NO	2	84.66	PASS R YES / E VALID
4	gpt-6.1-sol	ontology / r2	TIMEOUT	NR	1	60.04	NO_COMMIT_OBSERVED R YES / E NOT_RUN
5	gpt-5.6-sol	general_rule / r1	SUCCESS	YES	4	41.09	PASS R YES / E VALID
6	gpt-5.6-sol	general_rule / r2	SUCCESS	YES	4	39.39	PASS R YES / E VALID
7	gpt-5.6-sol	ontology / r1	SUCCESS	YES	4	36.68	PASS R YES / E VALID
8	gpt-5.6-sol	ontology / r2	SUCCESS	YES	4	29.55	PASS R YES / E VALID

Endpoint outcomes, receipt-derived host calls, and dispatch-to-terminal elapsed time.

Typed-intent proposal; deterministic planner/checker and Godot commit retain authority.

No efficacy/superiority/noninferiority/player-benefit claim; provider metadata and request counts remain NR.

Fig. 8. Native fixed-quest pilot from recorded traces: one authored quest, 8 assigned rows, 7 observed executions. Provider identity, revision, cost, and request count remain unknown. The assigned denominator retains one pre-response harness-serializer failure and two ontology response timeouts under the frozen 60 s `gpt-6.1-sol` deadline, separately from goal successes. This is a trace visualization, not a gameplay screenshot.

APPENDIX B
IDENTITY-REQUIRED EXTENSION

The admission gate in `research/runtime-model-swap-20261024/codex-identity-20261026/public-results-v1/results.json` (Table XX) requires response-bound provider model attestation (`openai-model`) before admitting an episode to a cross-selector comparison.

Identity-admission stop: not a model comparison. The separate extension has 20 planned rows, 1 assigned row, 1 native process, 0 identity-qualified rows and 19 unattempted rows. The first requested `gpt-6-luna`/`general_rule` CLI turn completed, but its response lacked provider model-identity evidence; it was not accepted and the batch stopped.

A requested selector or catalog advertisement is not a served identity. The evidence cannot distinguish an absent `openai-model` header from one not delivered as a valid string. This is not an observed model mismatch or model-incapability result. `gpt-6-terra` was not advertised by the pinned catalog, was not assigned and was not substituted.

Audit distinction: `record hash valid=true`; `engineering record accepted=false`. The original bootstrap label `episode_record_integrity_invalid` is retained. The recomputed hash matches; the compound predicate also requires engineering/safety PASS, so this label is not evidence of corrupted bytes. There were 0 commits, safety is `NO_COMMIT_OBSERVED`, and fresh-engine replay is `NOT_RUN`, not a safety pass.

Two blocked repair requests were not native reruns. The original eight-row pilot remains byte-identical and unpooled. Provider identity, revision, effective effort, request count, cost, actual credit use and cash charge remain `unknown`. Matching accounts and automatic reload OFF were observed before and after, without a setting change or purchase; these are not zero-cost or zero-credit attestations. No verified new model comparison, ontology effect or confirmatory A/C evidence follows.

Preparation 7.144 s; episode elapsed 4.257 s; native process 4.207 s; orchestration 11.566 s. These are distinct failed-attempt boundaries, not pure inference latency or time to success.

TABLE XIX
TEN IDENTITY-ADMISSION CELLS DISTINGUISH PLANS, ASSIGNMENTS, QUALIFIED IDENTITY AND UNATTEMPTED ROWS; NOT MODEL PERFORMANCE.

Requested selector	Representation	Planned	Assigned	ID-qualified	Not run
gpt-6-luna	general_rule	2	1	0	1
gpt-6-luna	ontology	2	0	0	2
gpt-6-sol	general_rule	2	0	0	2
gpt-6-sol	ontology	2	0	0	2
gpt-5.6-luna	general_rule	2	0	0	2
gpt-5.6-luna	ontology	2	0	0	2
gpt-5.6-sol	general_rule	2	0	0	2
gpt-5.6-sol	ontology	2	0	0	2
gpt-5.6-terra	general_rule	2	0	0	2
gpt-5.6-terra	ontology	2	0	0	2
Total	—	20	1	0	19

APPENDIX C
WITHDRAWN OLLAMA HISTORICAL ATTRIBUTION

The earlier local Ollama quantitative results, eligibility claims, and engineering-pass attribution are withdrawn because their original episode and response receipts could not be recovered. `research/maintenance/ollama-unverified-history-20261113.json` preserves the former section and table solely as unverified historical attribution withdrawn from current quantitative evidence, rather than original raw data or a validated experiment. No substitute observations, replacement receipts, or model recollections were produced, and the missing originals remain an explicit limitation rather than a passing or silently skipped verification requirement.

APPENDIX D
DIAGNOSTIC, HISTORICAL, AND BLOCKED COHORTS

A. Exploratory Signal Example

The exploratory example is an unverified historical attribution whose original receipt is unavailable and which falls outside the 19-cohort inventory; it remains excluded from active denominators and confirmatory evidence. That historical account describes a single synthetic `missing_effect` fixture omitting `signal_restored`, attributing objective success after one repair to a deterministic control. Under the requested `gpt-6-luna` selector, it reports an accepted repair and objective success under Detailed feedback, contrasted with empty effects, rejection, and unchanged-state fallback under Generic feedback (`Action rejected.`). While these attributed outcomes clarify the intended contrast in Fig. 6, they are not independently verified observations, and the available artifacts do not establish complete prompts, executed source-version bindings, or served-provider metadata.

B. Incomplete Synthetic Development Pilots and Bookkeeping

`research/codex_feedback_pilot/public-history-20261111/cohort-inventory.json` (Table XX) preserves assignment accounting and raw-file SHA-256 digests for 19 entries comprising 17 archived cohorts and two v5 selector collections. Earlier v2 evidence and v2 diagnostic cohorts for `gpt-5.6-luna` and `gpt-5.6-terra` (10 planned assignments each) either stopped at their first `missing_precondition` assignment with `protocol_error` (`FAILED`) or remained `IN_PROGRESS` or pending with 0 valid responses, with raw CLI error text omitted to prevent credential leakage. Across the 17 archived cohorts, the completed, failed, and pending counts are: original pilot $0/7/3$ (seven technical or interrupted attempts); four initial selector and native files $0/0/40$; initial v2 pair $0/0/20$; startup diagnostics $0/2/18$; v3 pair $0/0/20$; v4 Luna $10/0/0$ (detailed goal $4/5$, generic goal $4/5$) and Terra $0/1/9$; v2 evidence ($0/1/9$ per selector); and the v2 diagnostic snapshots noted above. The archive is therefore not uniformly aborted, and none of these 17 archived records is pooled with v5 in Table XVII.

C. Pre-Registered Fail-Closed Blocking Gate and Zero-Leakage Check

The pre-registered `research/prospective-pilot-20261103/protocol-v1.json` and `results-v1/` specify three selectors (Codex `gpt-5.6-sol`, Codex `gpt-6.1-sol`, Antigravity `gemini-3.8-flash-medium`; nominal medium effort) in two tracks (Table XX). Runtime crosses 3 targets $\times$ 2 information-matched representations $\times$ 10 fresh resets (60 units, 180 s per episode), and bounded development crosses 3 targets $\times$ 2 exploration modes (minimal input vs. repository exploration) $\times$ 2 representations $\times$ 5 tasks $\times$ 2 fresh runs (120 units, 900 s per assignment). The primary endpoint is budget-bounded validated success, with hard safety as a separate gate.

Prospective pilot status: not a model comparison. The frozen study `prospective-pilot-20261103` plans 180 units (60 runtime and 120 development) over three requested targets. 180 are blocked, 0 assigned, 0 attempted and 0 executed, with 0 native process starts. Status `NOT_RUN/BLOCKED`; scientific conclusion `NOT_MET`.

Success rate: null. Observed cost: null (known subtotal null; 0 admitted units with unknown cost). Effect estimate: null. A null is an absent measurement, never zero.

Blocked prerequisites, counted per unit: `BLOCKED_ACTIVATION_ABSENT` (180), `BLOCKED_IDENTITY_FORMAT_UNOBSERVED` (180), `BLOCKED_ISOLATION_UNPROVEN` (180), `BLOCKED_LIVE_PATH_NOT_IMPLEMENTED` (180), `BLOCKED_NO_CREDIT_ENFORCEMENT` (180), `BLOCKED_PUBLIC_DERIVED_UNAPPROVED` (180), `BLOCKED_SCHEMA_ACCEPTANCE_UNOBSERVED` (180), `BLOCKED_TARGET_UNVERIFIED` (60), `BLOCKED_TASK_FREEZE_INCOMPLETE` (180).

The earlier eight-row 5/8 pilot, the stopped identity extension, the Ollama lane, the deterministic fixtures and the n=1 results are separate lanes. They are neither pooled nor re-reported here.

Nine fail-closed admission gates blocked all 180 units before any model process launched. Subsequent independent offline controls verified acceptance of two correct variants and rejection of the defective baseline and plausible-wrong variant for all five development tasks, and the corpus leakage audit passed within its declared scope over 9,303 repository files. These checks establish pre-collection gating, fixture readiness, and leakage-check conformance rather than an executed pilot or model comparison, and no blocked run was fabricated or retroactively admitted.

APPENDIX E
EVIDENCE ARTIFACT AND DIRECTORY REFERENCE

Table XX indexes canonical paths, recorded fields, and verification entry points for contributions (**C1–C5**), active and historical lanes (**E1–E3**, **ENG1**, **NPC-ENG1/SCR1**, **NPC-DISC-1**, **DEV-CASE**), and the runtime and development cohorts.

TABLE XX
CANONICAL EVIDENCE DIRECTORIES, FILES, RECORDED FIELDS, AND VERIFICATION SCRIPTS FOR CLAIMS, LANES, AND APPENDIX COHORTS.

Lane / Scope	Directory and Key Files	Core Contents and Recorded Evidence Fields	Verifier & Reference
C1–C5 Core Contracts & ρ	`src/nesy_game/contracts.py`, `validator.py`, `repair.py`, `runtime.py`, `experiment.py`; `game-track/schemas/*.schema.json`	`WorldState`, `ActionPolicy`, 7 predicate checks (`VALIDATOR_CHECKS`), guided repair $\rho(a, E)$, state-reading oracle, semantic replay & SHA-256 chain verifier, Draft 2020-12 bridge/record schemas.	`scripts/validate_project.py` Sec. IV, Tables II–IV, Fig. 3
E1 Offline Conformance	`research/academic-pipeline/stage-04-pilot/pilot-results.json`, `pilot-assignment-manifest.json`, `sha256-manifest.json`, `*.csv`	13 gate fixtures (1 valid, 12 negative), 12 repair cases across 4 arms ($0/12$ reject/blind, $5/12$ ρ, $6/12$ oracle), 10 integrity-fault checks, 7 adapter terminal records, 3 manifest guards, 2 open sentinels.	`scripts/run_conformance_pilot.py` Sec. V, VI, Tables VII–VIII
Boundary Audit	`research/academic-pipeline/gemini-review-20261113/results-v2/`; `summary.json`, `rows.jsonl`, `surface_rows.jsonl`, `assignment_inventory.json`, `input_hashes.json`, `source_hashes.json`	1,664 main assignments and 48 separately denominated surface assignments; bounded authored finite-graph audit, frozen inventory, and input/source hashes.	`scripts/run_gemini_review_evaluation.py` `tests/test_gemini_review_publication.py` Secs. V-B, VI-D
E2 Archived Live Screening	`research/academic-pipeline/rq2-live-pilot/promotion-manifest.json`, `frozen-pilot-base/`, `signal-repair-v2/`	14 hash-pinned historical JSON/JSONL receipts (5 seeds $\times$ $K{=}1$ paired ρ vs. blind retry, $15/15$ non-commit state isolation); excluded from active claims.	`scripts/validate_contribution_crosswalk.py` Sec. V (Historical only)
E3 KG/Ontology Simulation	`knowledge/ontology/trace-rpg-ontology.json`, `curated-relations.json`; `research/simulation/kg-ontology/latest/evaluation-matrix.json`	43 OKF nodes, 106 reference edges, 24 curated typed edges; 6 edge-withheld queries $\times$ 7 fixed strategies (210 scores; `S2` $P{=}R{=}1.000$, MRR$=0.944$, Sem@3$=1.000$).	`scripts/run_kg_ontology_simulation.py` Sec. V, Table VI
ENG1 & **DEV-CASE**	`game-track/godot/project.godot`; `game-track/godot/docs/latest/evaluation-matrix.json`, `game-track/godot/docs/latest/balance-archetypes.json`; `game-track/web/dashboard/ontology-map.json`; `research/development-cases/ontology-map-20261023-01/` (`contract.json`, `result.json`, `evidence/verification-009-mobile-complete-tests.json`)	Retained matrix: 4 fixtures, 52/52 core and separate 4/4 extension checks; 5 archetypes. Eight defined smoke predicates are a contract, not a stored run-pass result. Bilingual 8-role map; $n{=}1$ development case with dated 62/62 test receipt.	`scripts/validate_game_track.sh`; `tests/test_godot_experimental_game.py` Sec. VI–VI-G, Table III
NPC-ENG1 & **NPC-SCR1**	`configs/npc-goal-recovery.json`; `research/academic-pipeline/npc-goal-recovery/verified-run-006/aggregate.json`, `*-results.jsonl`	16 deterministic `NPC-ENG1` cases and 72 `NPC-SCR1` rows ($P/R/O{=}24$ each), 12 stale-control ACK rejections (`SCR-T01/T02`), epoch-local replay, and pinned `fa9fb8a` provenance hashes.	`scripts/validate_npc_goal_evidence.py` Sec. V, VI, Table XII
NPC-DISC-1 Disclosure Gate	`src/nesy_game/disclosure.py`; `research/npc-disclosure/reveal-hint-expectations-v1.json`, `results-signal-explainer-20261023-r5/`	60 declaration-derived `reveal_hint` cases (48 normal$/4$ boundary$/8$ malformed) across 3 arms ($60/60$ `general_rule` vs. `ontology_gate` parity; 51 unsafe in unchecked arm) and 2 rule-change fixtures.	`scripts/generate_disclosure_paper_results.py` Sec. IV-B1, VI, Tables XIII–XV, Fig. 4
Native Pilot & **ID Extension**	`research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json`; `codex-identity-20261026/public-results-v1/results.json`	8 assigned fixed-game rows (5/8 goals, 1 harness fault, 2 timeouts, 23 host calls, 0/16 unsafe, 7 semantic replays, $6/0/1/1$ fresh-engine); plus planned 20-row identity extension stopped at Row 1 (`IDENTITY_UNVERIFIED`).	`scripts/generate_runtime_comparison_publication.py` Sec. VI, Table XVI, App. A–B
v5 Feedback & **Cohort History**	`research/codex_feedback_pilot/public-v5/`; `paired-model-comparison-20261110-v5-source.zip`; `public-history-20261111/cohort-inventory.json`	20 v5 assignments (5 fixtures $\times$ 2 selectors $\times$ Detailed/Generic; 8 gate pass, 7 goal, 10 schema fail), prompt/response SHA-256 digests, 7 frozen source files, and 19-cohort historical inventory.	`tests/test_codex_feedback_publication.py` Sec. VI, Table XVII, Figs. 6–7, App. D
Withdrawn History & **Blocked Pilot**	`research/maintenance/ollama-unverified-history-20261113.json`; `research/prospective-pilot-20261103/protocol-v1.json`, `research/prospective-pilot-20261103/results-v1/collection-status.json`, `summary.json`	Ollama attribution withdrawn: original receipts unavailable; historical JSON is not validated experimental data. Separate prospective plan: 0/180 completed (60 runtime$/120$ dev planned), 9 fail-closed blockers (`NOT_RUN/BLOCKED`).	`scripts/run_prospective_pilot.py` App. C–D
Crosswalk & **Manifests**	`research/academic-pipeline/*-matrix.csv`, `stage-05-citation-verification.json`; `visuals/source-manifest.json`; `research/traceability-scoring-20261023/rubric.json`	58-reference topic/status crosswalk (48 verified, 10 preprint), C1–C5 & E1–ENG1 claim-boundary matrices, visual/table SHA-256 source manifest, and 6-dimension traceability rubric (`TR01--TR06`).	`scripts/validate_visual_assets.py` Table I, Table V, Supplement